

Agile & DevOps Planning Document

Product Vision

Build a command-line calculator that performs arithmetic operations reliably, validates user input, and demonstrates Agile and DevOps practices through testing, CI/CD, and logging.

Definition of Done (DoD)

- Code implemented and functional
- Acceptance criteria satisfied
- Unit tests written and passing
- Code committed with meaningful history
- CI pipeline passes successfully
- Basic documentation updated

Product Backlog

US1 - Addition (2 pts, High)

As a user, I want to add two numbers so that I can get their sum.

Acceptance Criteria:

- User can input two numbers
- Correct result is returned
- Handles invalid input

US2 - Subtraction (2 pts, High)

As a user, I want to subtract two numbers so that I can get the difference.

Acceptance Criteria:

- User inputs two numbers
- Correct result returned
- Handles invalid input

US3 - Multiplication (2 pts, Medium)

As a user, I want to multiply two numbers so that I can get the product.

Acceptance Criteria:

- Correct multiplication result
- Handles invalid input

US4 - Division (3 pts, High)

As a user, I want to divide two numbers so that I can get the quotient safely.

Acceptance Criteria:

- Correct division result
- Division by zero handled
- Handles invalid input

US5 - Input Validation (3 pts, High)

As a user, I want invalid input rejected so the program does not crash.

Acceptance Criteria:

- Detect non-numeric input
- Show clear error
- Program continues safely

Sprint 1 Plan

Goal: Deliver basic calculator functionality with validation and CI setup.

- US1 Addition (2 pts)
- US2 Subtraction (2 pts)
- US5 Input Validation (3 pts)

Sprint Backlog:

- Setup project structure
- Implement addition
- Implement subtraction
- Add input validation

Sprint 2 Plan

Goal: Improve system with more operations, logging, and persistence.

- US3 Multiplication (2 pts)
- US4 Division (3 pts)

Sprint Backlog:

- Implement multiplication
- Implement division with zero handling